

JoLT: Joint Latent Trajectories for Context-Guided High-Resolution Tiled Generation

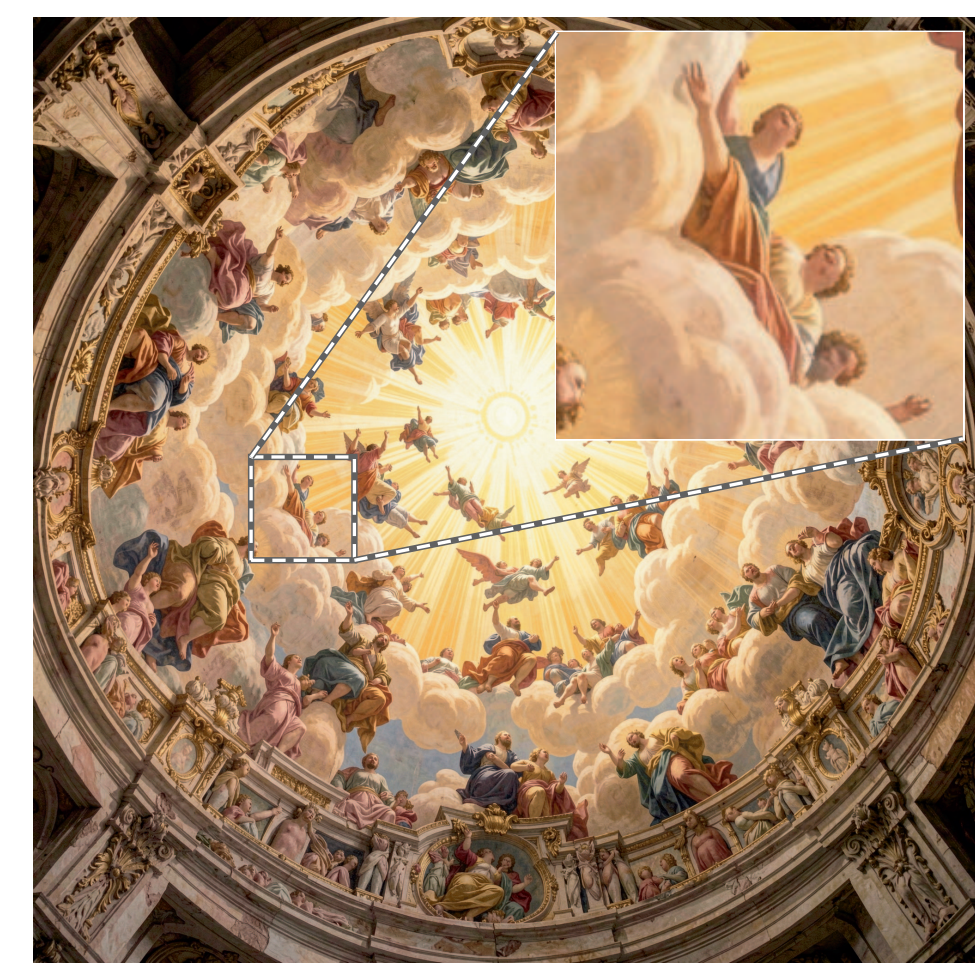
Mathis Koroglu^{1,2} Guillaume Jeanneret² Hugo Caselles-Dupré¹ Matthieu Cord^{2,3} Arnaud Dapogny²

¹Obvious Research ²Sorbonne Université, CNRS, ISIR ³Valeo.ai Paris, France

1 Motivation & Task

- **Goal:** Generate globally coherent high-resolution images with dense local meaning.
- **Complex image:** the density of semantic information is high locally, yet the scene is globally understandable; texture, repetition, or sharpness alone do not count.
- **JoLT:** global structure and local detail shape each other throughout generation.
- **Controllability:** explicitly steer visual complexity, style, and local content without fine-tuning.

Comparison, same prompt, added details



FLUX.2 Klein-9B
2048²

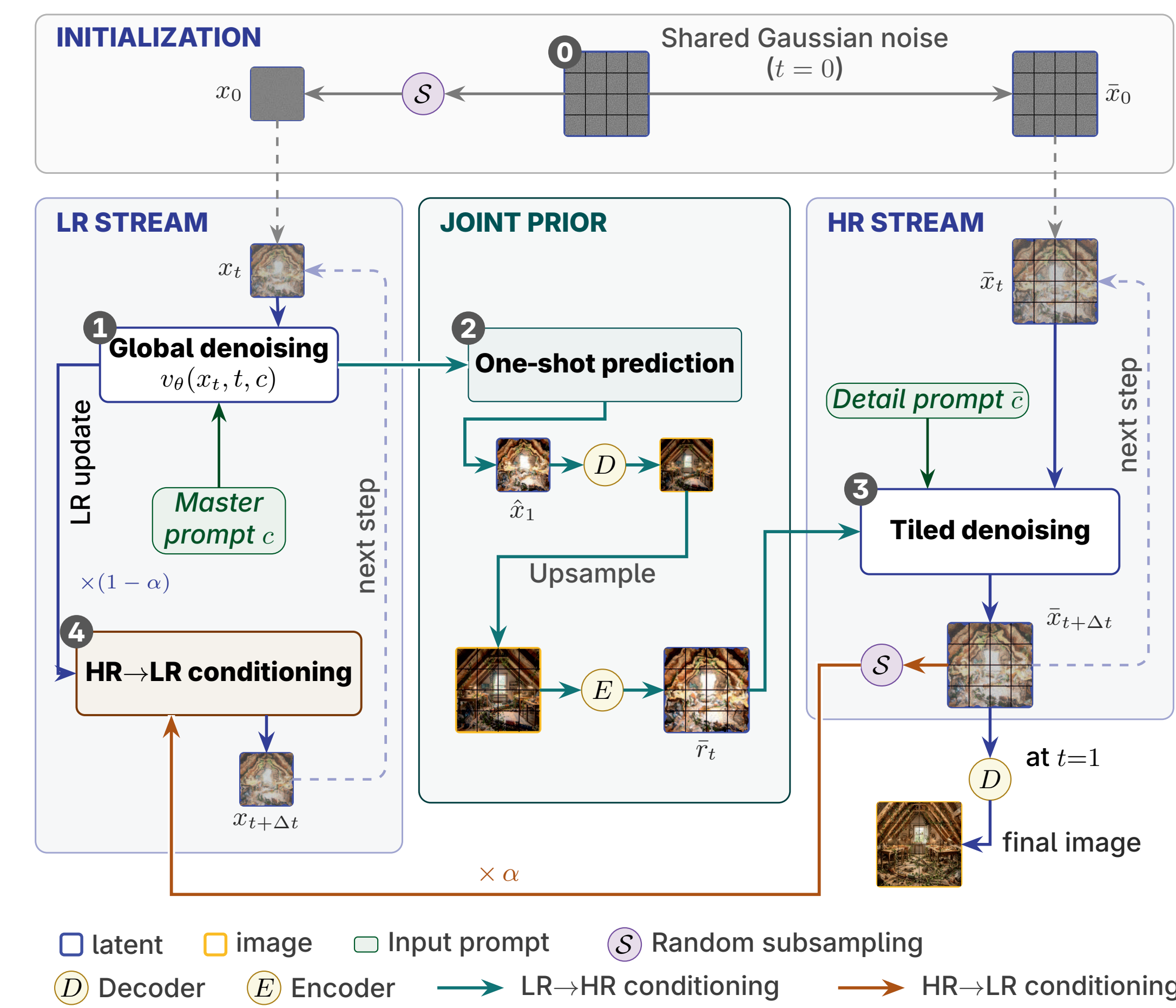


JoLT
4096²

Our detail prompt adds coherent local structure at 4K res.

2 Method

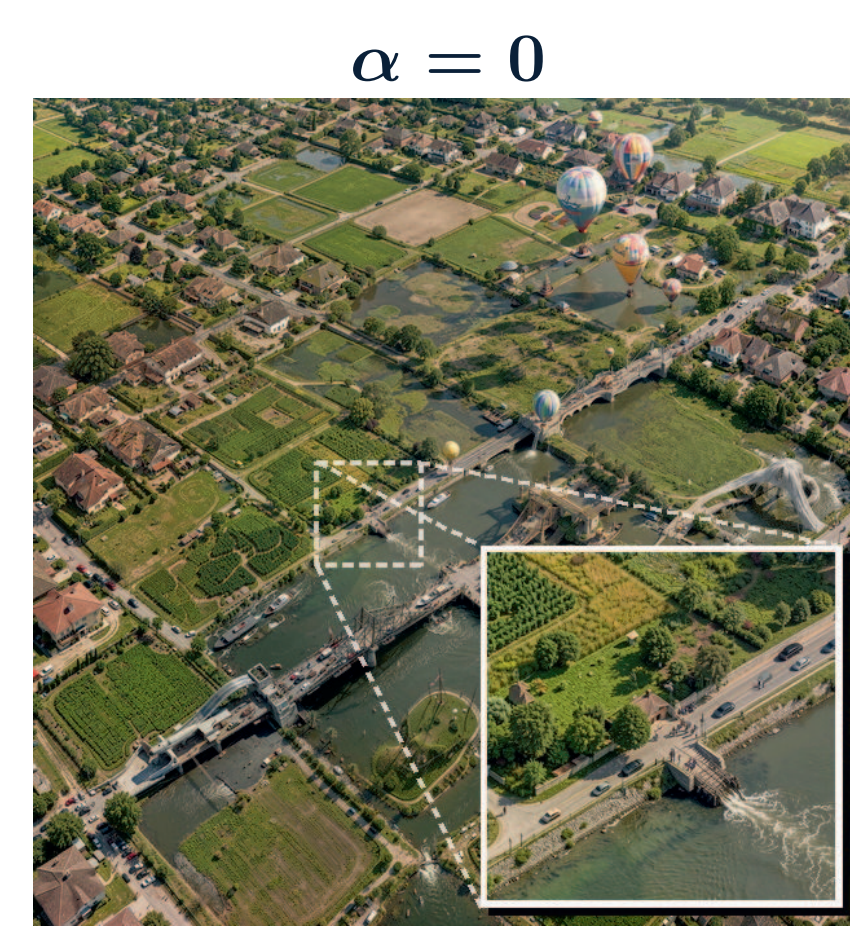
JoLT couples global low-resolution (LR) and tiled high-resolution (HR) trajectories. Prompt c sets the scene; detail prompt $\bar{c}$ controls added fine elements.



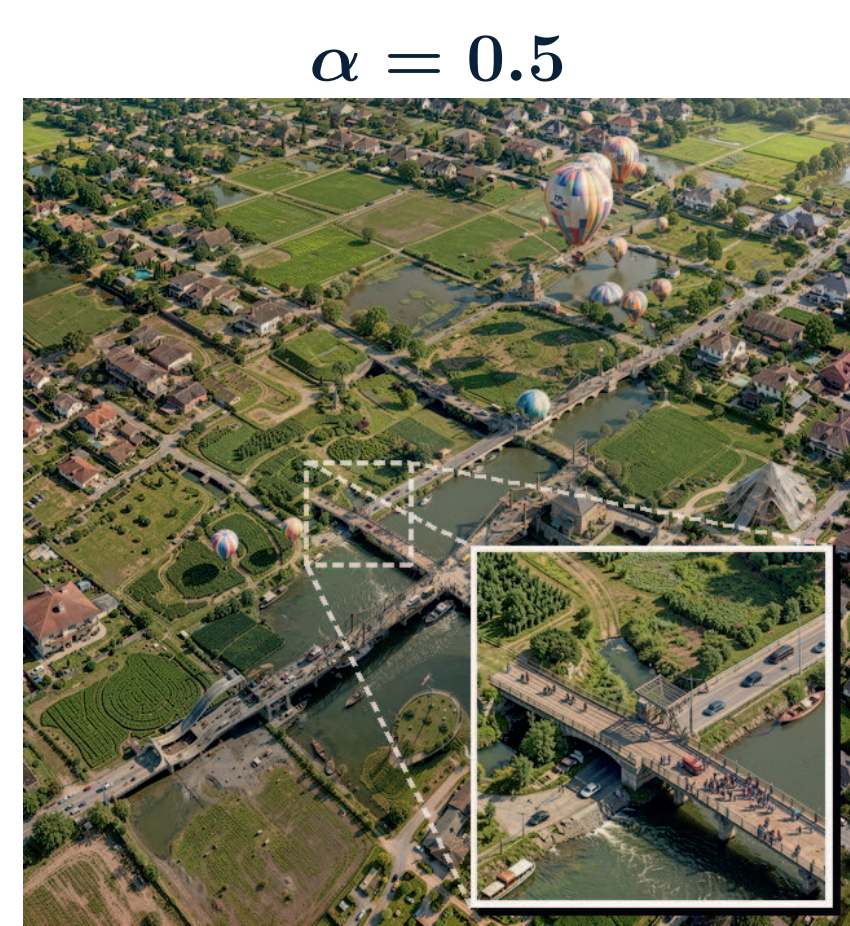
- **LR $\rightarrow$ HR:** the joint prior guides tiled layouts.
- **HR $\rightarrow$ LR:** we inject HR into LR by random subsampling and latent merging, with controllable $\alpha \in [0, 1]$.

Cost: mean 127 s per 4K image on one H100 (four steps, bfloat16, no training). Baseline means: 75–125 s.

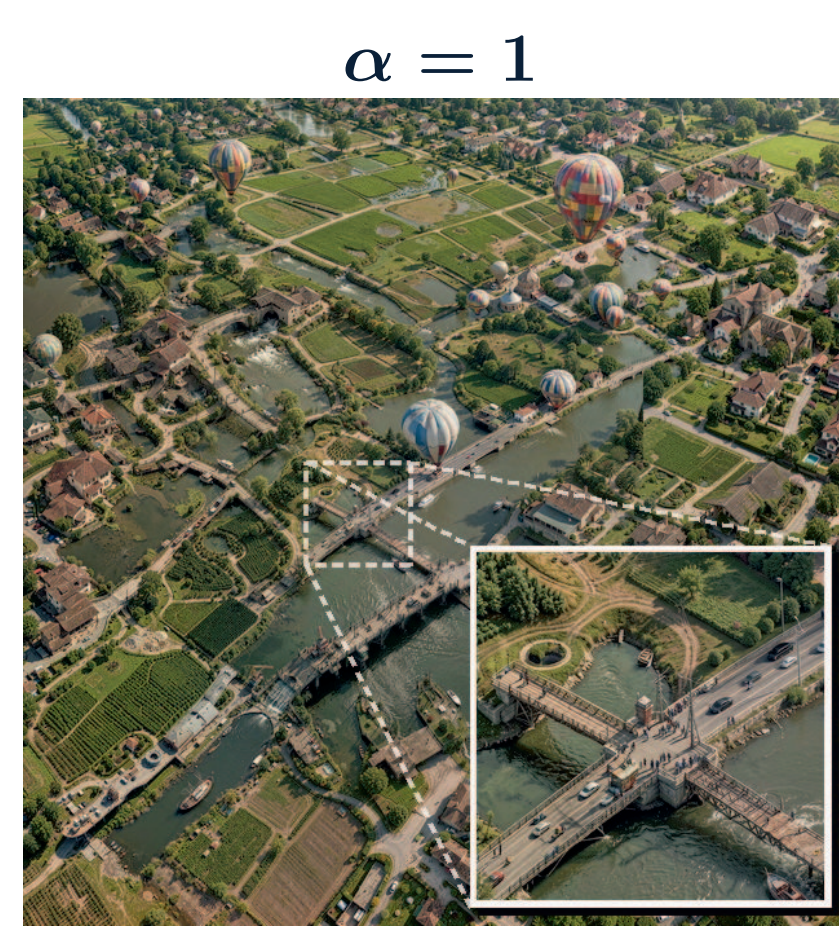
3 α sweep: apparent complexity



$\alpha = 0$



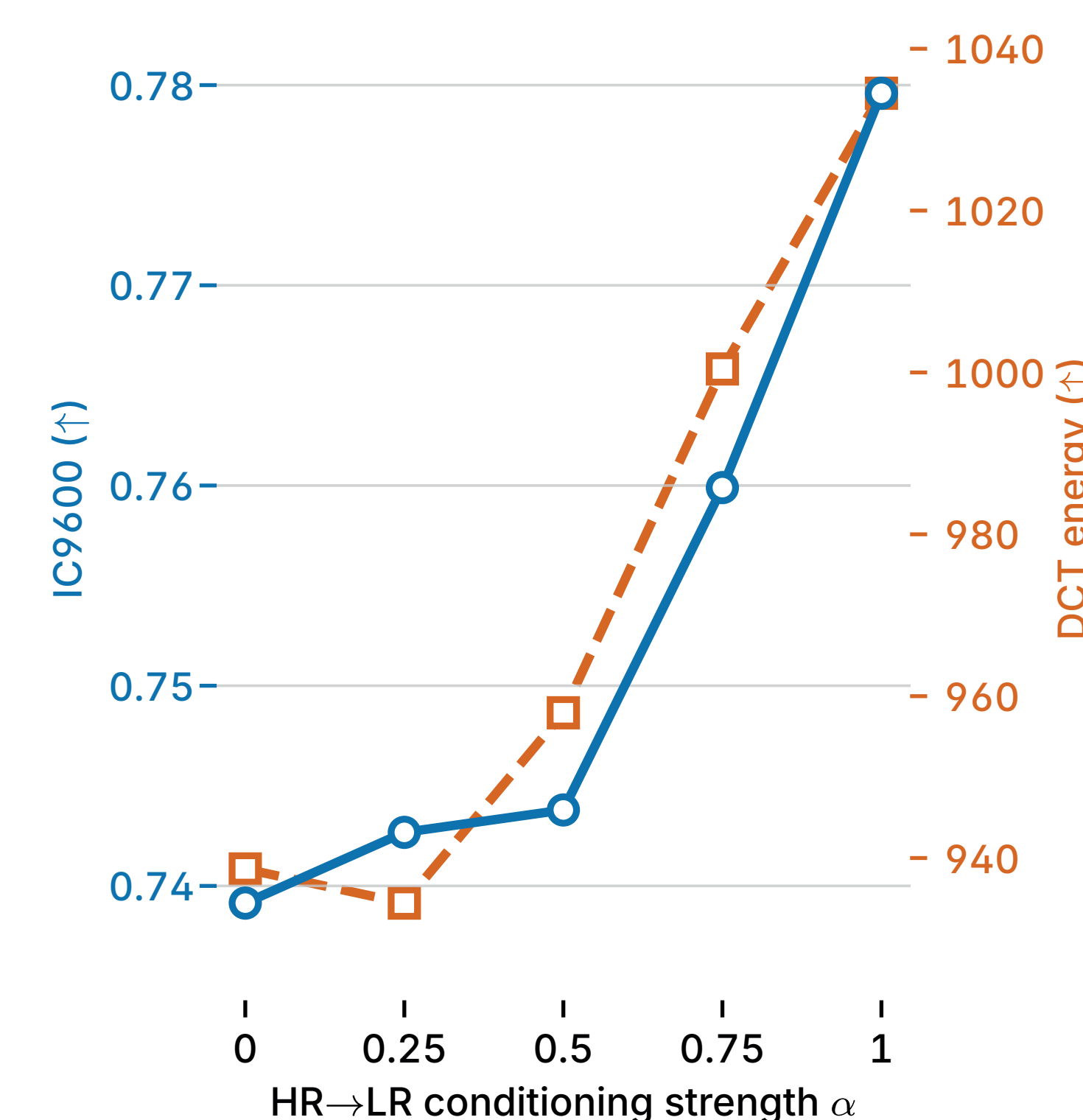
$\alpha = 0.5$



$\alpha = 1$

Prompt: Aerial flooded suburb [...] Three α values; insets show local detail.

Complexity metrics. IC9600 estimates perceived complexity; discrete cosine transform (DCT) energy measures high-frequency structure. Higher means more complex, not better.



- **Feedback strength.** α controls how strongly HR feeds back into the LR trajectory; its effect depends on the detail prompt $\bar{c}$, not on α alone. At $\alpha = 0$, LR ignores HR, while HR still receives the LR prior.
- **In this sweep,** apparent complexity rises with α ; prompt alignment stays stable.

4 Results



Same setup: all baselines are reimplemented on FLUX.2 Klein-9B at 4K with four steps across 200 prompts. **Check the paper for quantitative metrics.**

User preference for JoLT

vs.	More detailed	Better match	More artistic
SEGA	77.3 $\pm$ 5.1%	69.0 $\pm$ 5.4%	68.0 $\pm$ 5.5%
DemoFusion	76.3 $\pm$ 5.1%	48.5 $\pm$ 5.6%	46.3 $\pm$ 5.7%
ResDiT	76.0 $\pm$ 5.1%	68.3 $\pm$ 5.5%	71.0 $\pm$ 5.4%

2,700 pairwise judgments over 120 prompts.

Takeaways

- **Joint:** composition and detail develop together.
- **Bidirectional:** LR and HR guide each other at every step.
- **Control:** α sets feedback strength
 $\bar{c}$ sets local content and style.

5 Ablations

Detail prompt: what kind of detail

- **It can change:** Texture, lighting, material, style, details, or objects.
- **Varying only $\bar{c}$:** Prompt c and seed stay fixed; the scene is preserved.



"Enhance the image, add lots of details"



"pointillist style"



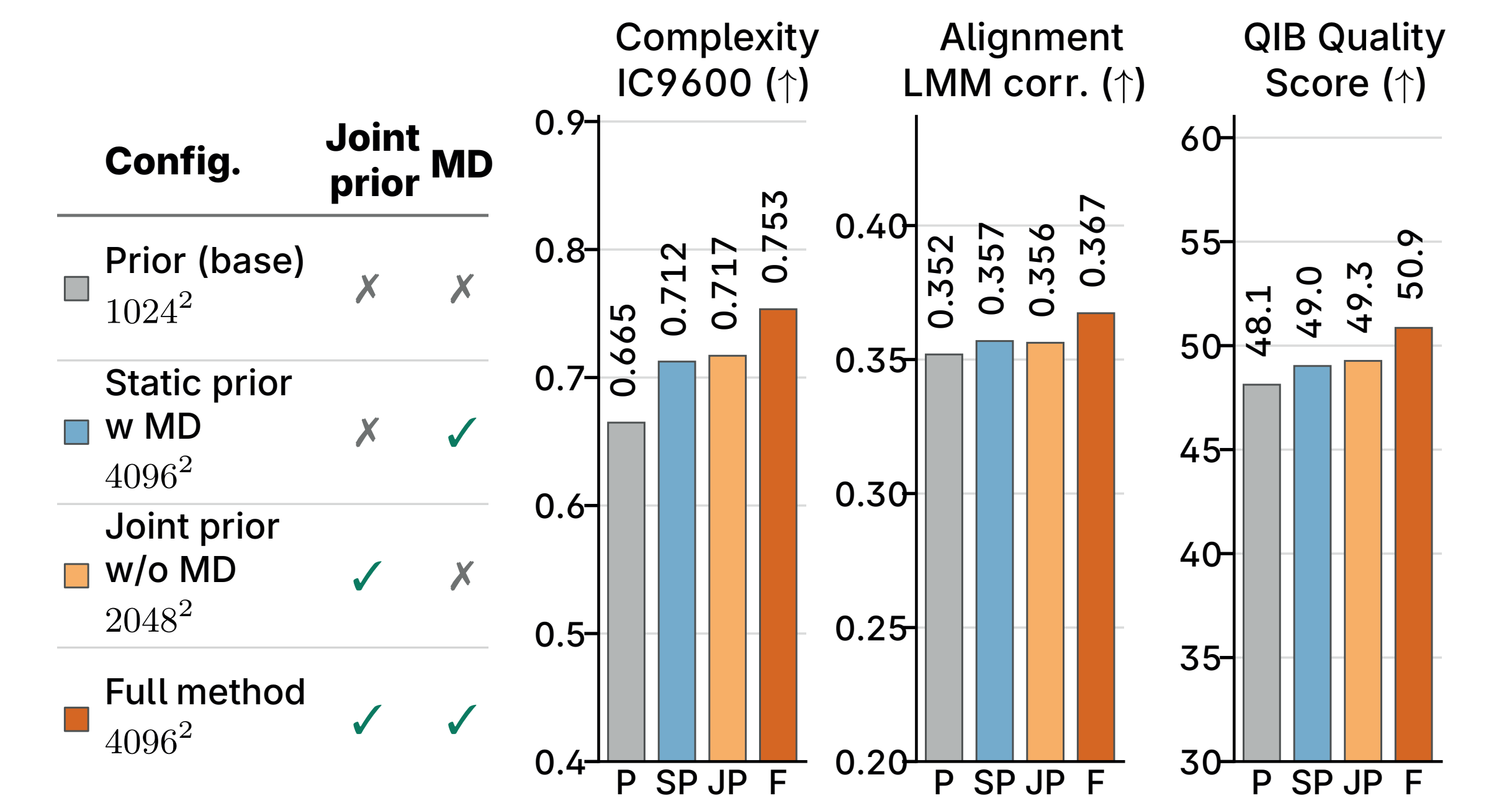
"Add iridescent caustic light reflections"



"Intricate grotesque imagery"

Which components matter?

Metrics. LMM corr. = LMM4LMM correspondence. QIB = Qwen-Image-Bench (Qwen 3.6 27B) Quality: realism, detail, resolution.



Four configurations isolate the LR prior, joint LR–HR denoising, and MD (MultiDiffusion); 200 prompts.

- **Joint vs. static, 4K:** IC9600 +5.7%; prompt alignment +2.9%; quality +3.7%.
- **With MD, 4K:** IC9600 +5.1%; prompt alignment +3.1%; quality +3.2%.